Protecting Salem's Ecosystem by Preventing the Spread of the Emerald Ash Borer

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Team Metadata

Team Members

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Studio Session: 3

Team Size: 3 people

Problem Framing and Research Question

Emerald Ash Borers are pests that are native to Northern China, Russia, Japan, and Korea (USDA, 2023). These insects are highly corrosive to North American Ash Trees, and are killing trees across the United States (PennState Extension, 2025). These beetles were found in Oregon for the first time in 2022 and have been steadily increasing their lifespan as the climate changes (Boboc, 2026). This recent expansion caused Oregon to start thinking critically about readiness and response plans, including Salem's own plan (City of Salem, 2026).

Due to the detrimental effects the Emerald Ash Borers have on tree canopies and general ecology in North America, many disciplines have become concerned and involved with finding solutions, including biologists, ecologists, land managers, engineers, and data scientists. There isn't much existing research that is publicly findable on the intersections between data science and Emerald Ash Borers, but some papers have been published about using Geographic Information Systems to survey land and predict what areas are most likely to become infested (Huset, 2013). These seem to have very limited geographic scopes, and to the best of our knowledge, there is no existing literature on data science analysis of Emerald Ash Borers in Salem, or in the Mid-Willamette Valley specifically.

Some concerns that are highlighted within the City of Salem's (2026) "Emerald Ash Borer Readiness and Response Plan" include riparian zone canopy reductions, climate-inspired increase in population of the Emerald Ash Borer, and threats to wildlife. Trees can become infested when they are healthy, but it is more likely if the tree has already suffered damage, such as drought or heat stress (City of Salem, 2026). Additionally, Oregon Ash trees, which are commonly found throughout Oregon, are highly susceptible to Emerald Ash Borers (Oregon Department of Forestry, 2024). Oregon Ash trees make up a portion of the canopy coverage in riparian zones; if those trees are lost in the future due to Emerald Ash Borer infestations, there will be gaps in the canopy above waterways, leading to increases in water temperatures.

With all of this information, we have deduced several research questions to apply over the course of this semester:

1. Can we predict which stands of trees Emerald Ash Borers are likely to infest in Salem, OR?
    a. Stand: "a contiguous group of trees relatively uniform in age-class distribution, composition and structure, and growing on a site of sufficiently uniform quality, that is a distinguishable unit from adjoining areas or stands" (Mercker and Hoyt, n.d.).

2. Can we model what the prevalence of trees in riparian zones will look like along Salem's waterways in the next 5 years?
    a. Riparian zones: the narrow strips of land that border creeks, rivers or other bodies of water (Oregon State University, n.d.).
3. Can we predict how stream temperatures will be affected by a loss of Oregon Ash trees and the shade provided by them?

Our vision for this project is to help government and private organizations prevent the spread of the Emerald Ash Borer and mitigate its impact on local ecosystems. By developing an interactive map and dashboard that models the beetle's effects on Salem's waterways, this project will identify at-risk areas and highlight targeted preventative measures that can be taken.

We are specifically interested in this project because the three of us grew up in the Willamette Valley, surrounded by nature and a curiosity for stewardship. This unification brings a personal element to the heart of the project.

Impact and Stakeholders

The results of this project will benefit the City of Salem's Department of Natural Resources, which is a stakeholder of the project. Using these results, a more informed plan can be created by the City of Salem to protect riparian zones in Salem, along with the indigenous species that live in them.

The Shared Theme domain for our project is the Pacific Northwest forest and environmental health. Our Peer POs are looking into fire patterns in Oregon, and could support different perspectives about topics such as tree weaknesses from drought or heat, as well as heat ecology damages in general that may compound our data. This will help us think holistically about the project.

Data Sources and Access Plan

Data Source	Brief Description	Access	Volume/Cadence	Instructor Approval Status
Emerald Ash Borer Web Map [1] And it's associated data [1 a]	This map provides the latest information on survey data and areas at risk for emerald ash borer (EAB) spread and ash tree mortality. General management recommendations and quarantine restrictions are provided. Publicly Available.	Public website to be scraped	Management buffers and state quarantine boundaries are updated each year in the summer and fall, or after significant updates are necessary	Requested
Tree Plotter [2]	This is an inventory of all trees in Oregon. This would allow us to see where Oregon Ash Trees are located. Publicly Available.	Public data to be scraped	As published, estimated less than or equal to once a year.	Requested
Tree Plotter Canopy [3]	Very similar to above. Maps tree types and their locations. Only in the Salem area. Publicly Available.	Public data to be scraped, or can be downloaded using a stakeholder's login information	As published, estimated less than or equal to once a year.	Requested
Oregon Department of Forestry Land Cover Willamette	Very similar to the previous 2 data sources. Maps tree types and their locations. Publicly Available.	Public data to be scraped	As published, estimated less than or equal to once a year.	Requested

Valley Hardwood Model [4]				
Oregon EAB Trap App (PUBLIC) [5]	"ODF also coordinates a statewide trapping program for EAB with over 40 participating local agencies. The trapping season is just starting now. We have handed out over 500 traps to cooperators. They will be populating this map over the next 2-3 weeks and those updates will be live." Publicly Available.	Public data to be scraped	Updates are live, we could collect data weekly.	Requested
Stream/ Waterway	This data comes from the City of Salem and contains information at the 15 minute level for temperatures of streams in degrees Celsius. Each stream is contained in a different .csv file.	Publicly accessible via data requests sent to the City of Salem. We have already requested and received this.	Daily, with temperature data populating every 15 minutes	Requested
Emerald Ash Borer (EAB) in Oregon [6]	Oregon's website dedicated to the Emerald Ash Borer. Contains information about the bug itself, infestation spread prevention, and their impact on Oregon Ash Trees. More for general information than for data collection.	N/A	N/A	N/A

Links for datasets

1. https://experience.arcgis.com/experience/9f29b1860cb04d36ad71b122148277f3/page/Page/
    a. https://oregoninvasiveshotline.org/reports/list?q=Emerald+Ash+Borer&submit=Search&categories=6
2. https://pg-cloud.com/Oregon/

3. https://pg-cloud.com/SalemOR/
4. https://geo.maps.arcgis.com/apps/mapviewer/index.html?layers=565c6a9705724f0baead0c118aee9c92
5. https://geo.maps.arcgis.com/apps/mapviewer/index.html?webmap=cd7f5da6130749a9b7b99a45844b0f24
6. https://oregon-eab-geo.hub.arcgis.com/

Data Engineering Plan

Ingestion: The data will be ingested by downloading the data and using an API.

Stored: The data will be stored in the data/raw folder on the project repo.

Versioned:

- data/raw/aqs/: API pulls by county FIPS and parameter code.
- data/raw/places/: download CSV snapshots versioned by download date.
- data/interim/: cleaned data, but not engineered.
- data/processed/: analysis-ready panel for modeling.
- src/ingest/aqspull.py: reproducible AQS download script.
- notebooks/01_panel_qc.ipynb: row counts, missingness, duplicate checks.

Documented: Our data will be documented each time it is uploaded with the name, details of ingestion, and what changes or cleaning were done.

Planned Analysis

1. **Set up Railway:**
    a. **Create:** Railway Project for our capstone. This will allow us to ingest data from CSV files into a shared space and set up web scrapers. Also set up a connection to this project in our individual PgAdmin or Beekeeper.

b. **Observe**: N/A

c. **Analyze**: N/A

d. **Success:** We will have a functional Railway Project and successfully connect to it from each of our devices.

2. **Collect the data:**

   a. **Create:** We will download and scrape all of the data sources using an API.

   b. **Observe:** We will ensure data has been populated into the table and note the data structure for each.

   c. **Analyze:** We will look into the raw CSVs to look for obvious errors, such as duplicate rows or discrepancies

   d. **Success:** We have at least one data source on Emerald Ash Borer Prevalence, at least one on Tree types and locations, at least one on Oregon Ash Borer Trap Data with recurrence via API, and successfully collected the Stream/Waterway Data.

3. **Exploratory Data Analysis:**

   a. **Create:** We will create basic summaries of each dataset with descriptive statistics included.

   b. **Observe:** We will consider whether these summaries are relevant for inclusion in the final report.

   c. **Analyze:** We will check in with our peer POs for analysis of the data to confirm that the statistics are sensible.

   d. **Success:** We collect the mean, median, mode, and quartiles for every numeric column. We collect all possible values for qualitative columns, as well as the most common value for each column.

4. **Machine learning model for at-risk stands of trees (RQ1):**
    a. **Create:** We want to create a model which can predict the risk level of each different stand of trees in the Salem area. We will create a Decision Tree, a Random Forest, an XGBoost, and a Support Vector Classifier.
    b. **Observe:** We will pursue and fine-tune the model with the highest accuracy and cross-validation score.
    c. **Analyze:** We will use the last 2 years of data for the test set and the previous data for training. We will evaluate the models based on accuracy and cross-validation.
    d. **Success:** Achieved a model with an accuracy above 90% and a cross validation score above 75%.
5. **Machine learning model for tree prevalence along waterways for next 5 years (RQ2):**
    a. **Create:** We want to create a model which can predict the number of Oregon Ash Trees along waterways. We will create geographic boundaries which we will use to separate each area of prediction. We will create Linear Regression, Polynomial Regression, Random Forest Regression, Support Vector Regression, ElasticNet Regression, and Bayesian Linear Regression models. We will remove the last 2 years of data to test the models on.
    b. **Observe:** We will pursue and fine-tune the model with the lowest RMSE, lowest MAE, highest R squared, and highest cross validation score.
    c. **Analyze:** We will use the last 2 years of data for the test set and the previous data for training. We will evaluate the models based on RMSE, MAE, R squared, and cross validation score for predicting values for the next 5 years.

    d. **Success:** Achieved a model with a relatively low RMSE and MAE, an R squared above 75%, and a cross validation score above 75%.

6. **Regression model for negative impact on temperature by loss of trees (RQ3):** *Create:*
    a. **Create:** We will create a regression model that can predict the temperature of streams due to loss of trees. We will create Linear Regression, Polynomial Regression, Random Forest Regression, Support Vector Regression, ElasticNet Regression, and Bayesian Linear Regression models. We will remove the last 2 years of data to test the models on.
    b. **Observe:** We will pursue and fine-tune the model with the lowest RMSE, lowest MAE, highest R-squared, and highest cross-validation score.
    c. **Analyze:** We will use the last 2 years of data for the test set and the previous data for training. We will evaluate the models based on RMSE, MAE, R-squared, and cross-validation score for predicting values for the next 5 years.
    d. **Success:** Achieved a model with a relatively low RMSE and MAE, an R-squared above 75%, and a cross-validation score above 75%

Program Integration

We intend to use our knowledge in research design to guide our project plan, create and guide our data collection and data analysis methods. We have used these skills in previous classes, which have given us an understanding of how to safely collect public data and load it into a database. We have also had a lot of practice performing EDA, so we can leverage this experience to clean and analyze our data effectively. We will use our data engineering skills to bring all this data together in a single database (Postgres). Once this data is brought together and cleaned, we will be able to use our knowledge from the Machine Learning courses to test different models on our data to answer our research questions. We will also use general data summaries and analysis to explain our data and the context around it.

Our background in the Data Engineering course will also support our goal of creating a Grafana Dashboard to summarize our stream temp predictions and general data summaries. For any graphs used in the Grafana Dashboard, we will leverage our teachings from Data Visualization to create easy-to-read and informative visualizations. Finally, we will continue to use our skills from Data Visualizations when creating an interactive map to demonstrate the areas where Oregon Ash Trees are most at risk of the Emerald Ash Borer and where tree prevalence could see the greatest impacts.

Ethics and Risk Notes

It is always possible that our models are not accurate enough, and that could lead to protective measures being taken in the wrong areas, while areas in need of these protections are left at risk. We do not anticipate any issues with open data, as we have consent from the City of Salem to use this information publicly.

We anticipate our main risk for this project being a lack of time. We worry that gathering and collecting this data will be time-consuming, and we want to ensure that we are thorough to avoid little mistakes early on. To mitigate this, we will generally try to start tasks early enough that we can ask for help or fix errors early should they arise. Another potential risk is that we don’t have enough data to create a successful model. We feel that this is likely not going to be a problem, but we won’t know for sure until we do some EDA on the data and begin that process.

Timeline and Milestones

Week 4: Project Proposal Completed, Data Sources Identified

- We made it through these steps before writing this document and didn't run into any problems.

Week 5: Download all data, ingest it into Postgres, and start Railway

- Risks: Can we download the data? Is the data usable? Do we have access to the data? Do we have to use web scrapers to gather the data? Do the APIs work effectively? Can we remember how to start a Railway Project?
- Mitigation: Start early! We want to start exploring these as soon as possible so that we have time to deal with all these potential risks. Most of the data we have gotten from people working for the State or Oregon so we feel confident that we can make it work but we won't know until we try.

Week 6: Data Cleaning Complete and Final Data Sources Selected

- Risks: Do we understand the values in the data? Do our data sources allow us to answer our research questions?
- Mitigation: We will continue to reach out to our connections at the State of Oregon to understand the data if needed. And we will continue to stay ahead on this project so that we can continue to search for new data sources if we don't feel like the current ones can answer the research question. Make sure we have enough variables to train models on the data.

Week 7: Data Summary Completed and EDA/Feature Engineering

- Risks: We can't engineer any features. We are tight on time and won't get all the ingestion done on time.

- Mitigation: We follow this plan and really push to get the milestone done on time. We should always be able to engineer new features from the data if we understand it and the context of our subject well. If we keep the modeling in mind while bringing in the data it should keep us on track.

Week 8: Models Completed for RQ1, RQ2, and RQ3, Documentation and Process for Write-Up Completed. Rough outline of the interactive map and dashboard completed.

- Risks: With the data we have accessible, we cannot create models that answer our research questions successfully.
- Mitigation: In the earlier weeks of our project, we will keep the modeling steps in mind so that we can foresee if our data will not be able to create a successful model.

Week 9: Draft Complete of Interactive Map and Dashboard Design

- Risks: The interactive map could be very challenging or time-consuming to make. We might need multiple interactive maps to show data over time, which might be clunky.
- Mitigation: Reach out to Prof. Smalley if we need help on aspects of data visualization. Ask for help in general if we get stuck. Make sure we start these aspects early and create a plan in week 8 so we only have to implement it this week.

Week 10: Poster Rough Draft

- Risks: We could potentially not have enough information to answer our research questions to create a good draft.
- Mitigation: In the early stages of our modeling, we can have a sense of whether we will have enough information or not to create a comprehensive draft to answer our questions. We can explain where things went wrong, and if necessary, we could grab emergency data to make our project meaningful again.

Week 11: Finalize Map/Dashboard and Poster Setup, Begin Write-Up

- Risks: We could realize that our data doesn't actually support our research question or isn't actually answering the questions posed to us by our partners at the State or Oregon.
- Mitigation: Be able to explain why things went wrong. If really necessary, we could go back and gather emergency data to make our project meaningful again.

Week 12: Complete Write-Up Draft

- Risks: We don't have the project pieces finished so we can't complete parts of the write up.
- Mitigation: Any draft is still better than none. We could write about what we assume will happen when we finish those missing sections and tidy it up/ correct it for the final submission.

Week 13: Final Edit of Write Up and Practice Final Presentation

- Risks: Our write-up draft is not complete yet and it will take extra time to make it complete before editing. Our final presentation relies on our draft being complete so that we can effectively communicate our findings.
- Mitigation: We will make sure we are on time with the write-up so that we are not pressed for time in the editing stages, and when we need to create our presentation.

Week 14: Final Presentation

- Risks: Someone is ill. We generally don't feel prepared.
- Mitigation: We all still know each other's parts of the project so we aren't scrambling the morning of the presentation. We should be practicing the presentation ahead of time so that we know what to expect on presentation day.

DS3 Process and Tooling

- GitHub repo: https://github.com/Serenna4/Emerald-Ash-Borer-Capstone-Project
- GitHub Projects board: https://github.com/users/Serenna4/projects/2
- All team members have joined Discord and GitHub. We have also gained access to the projects for which we are Peer POs.
- The CHARTER.md document is completed, and a backlog is created.
- We have agreed to have our Studio Briefs due Sunday at 5 pm, so they are ready for the next day's class. The critiques are due Wednesday at midnight. And updated Backlogs / Iteration Review should be completed by Saturday at 5 pm so that the Briefs can be written on time.

References

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